Rearranging Neutral Atom Arrays for Quantum Computing

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Why Neutral Atom

Scalability: Neutral atoms scale more easily than other platforms.

Stability: Less sensitivity improves system reliability.

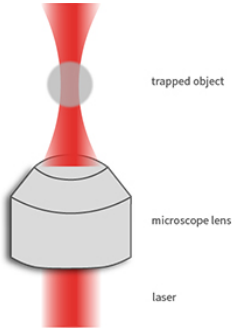
Control: Atoms can be precisely manipulated.

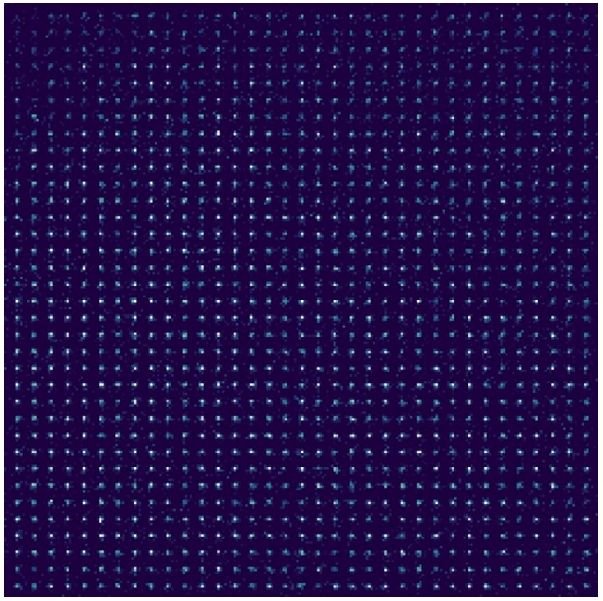
Arrays: Large qubit arrays are achievable.

Feasibility: High number of qubits become more realistic.

Experimental setup

- **Optical tweezers:** Deterministically trap single atoms.

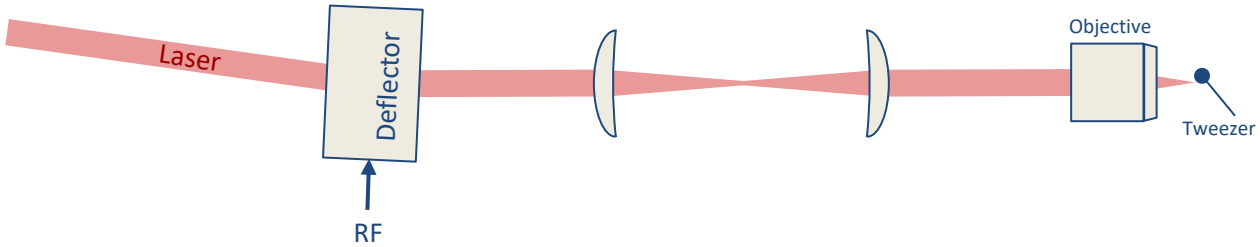




Neutral atoms array: approximately half filled sites

Experimental Setup:

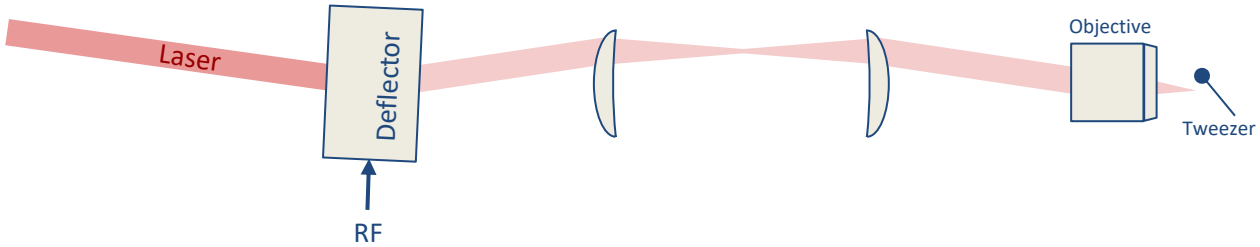
Laser → Beam steering → Tight focusing → Atom trapped



Dynamic Control: By sweeping the RF frequency, we can move the trapped atom in real-time.

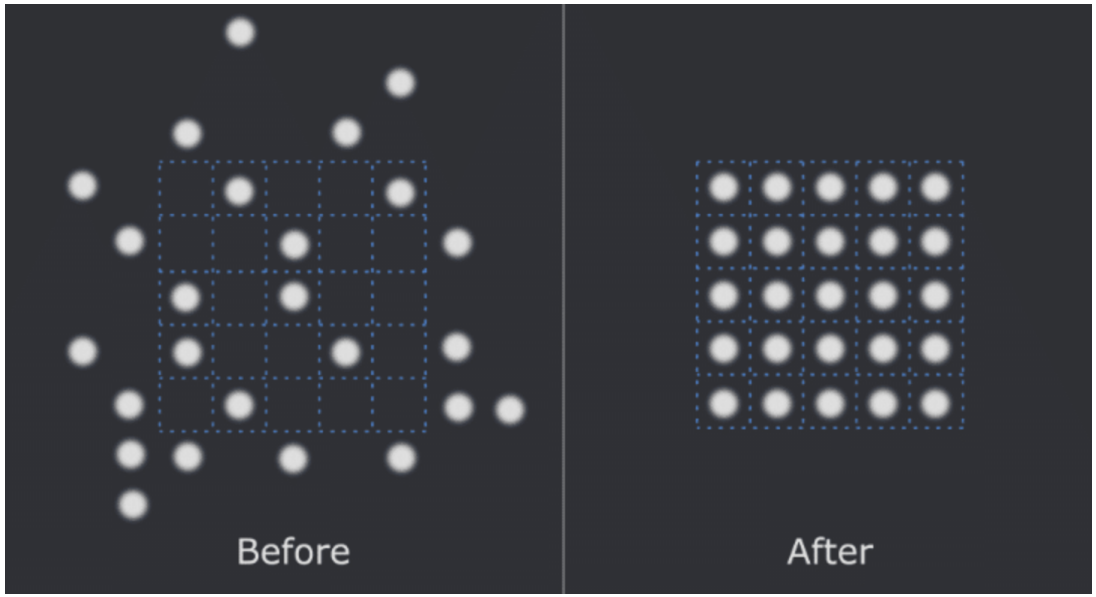
Experimental Setup:

Laser → Beam steering → Tight focusing → Atom trapped



Dynamic Control: By sweeping the RF frequency, we can move the trapped atom in real-time.

Challenge: Array Imperfections

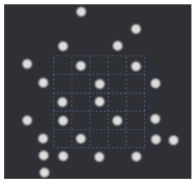


Source: https://www.quantum-machines.co/solutions/neutral-atoms/

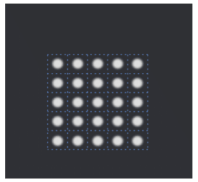
- Stochastic loading creates empty sites
- High-fidelity computation needs perfect arrays

Problem Statement:

- Given a random occupancy matrix



- Find the paths to fill the target sites



Optimization Problems:

- Assignment problem

  Find: $T = \{1, \ldots, n\}, \qquad A = \{1, \ldots, m\}$

  $$f : T \to A$$ (Injective)

- Scheduling problem
  Find optimal path for each assignment and the sequence of moves

Optimization Tools:

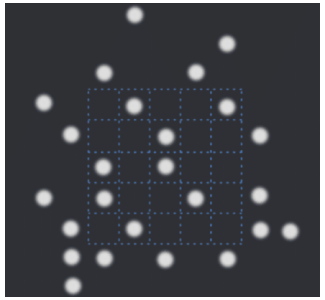
- Distance Matrix (Manhattan)

$$D_{ij} = |x_i - x_j| + |y_i - y_j|$$

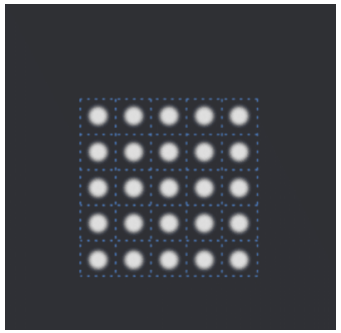
- Cost Function

$$C(f) = \sum_{i \in T} D_{i,f(i)}$$

Work Flow



Occupancy matrix



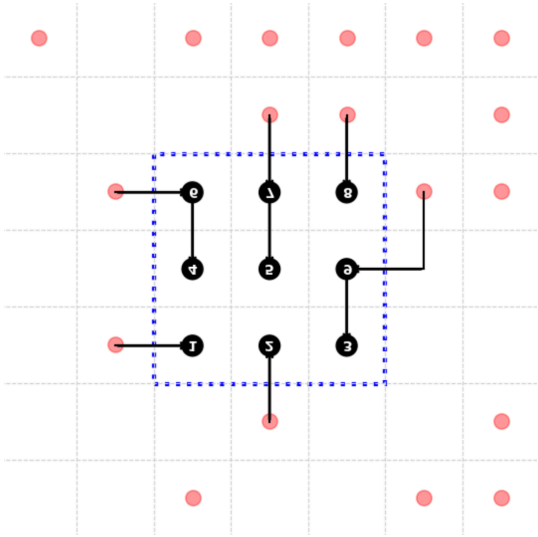
Target matrix

Algorithm

Brute-Force O(N!)

Heuristic Shortest Path O(N^3)

Hungarian

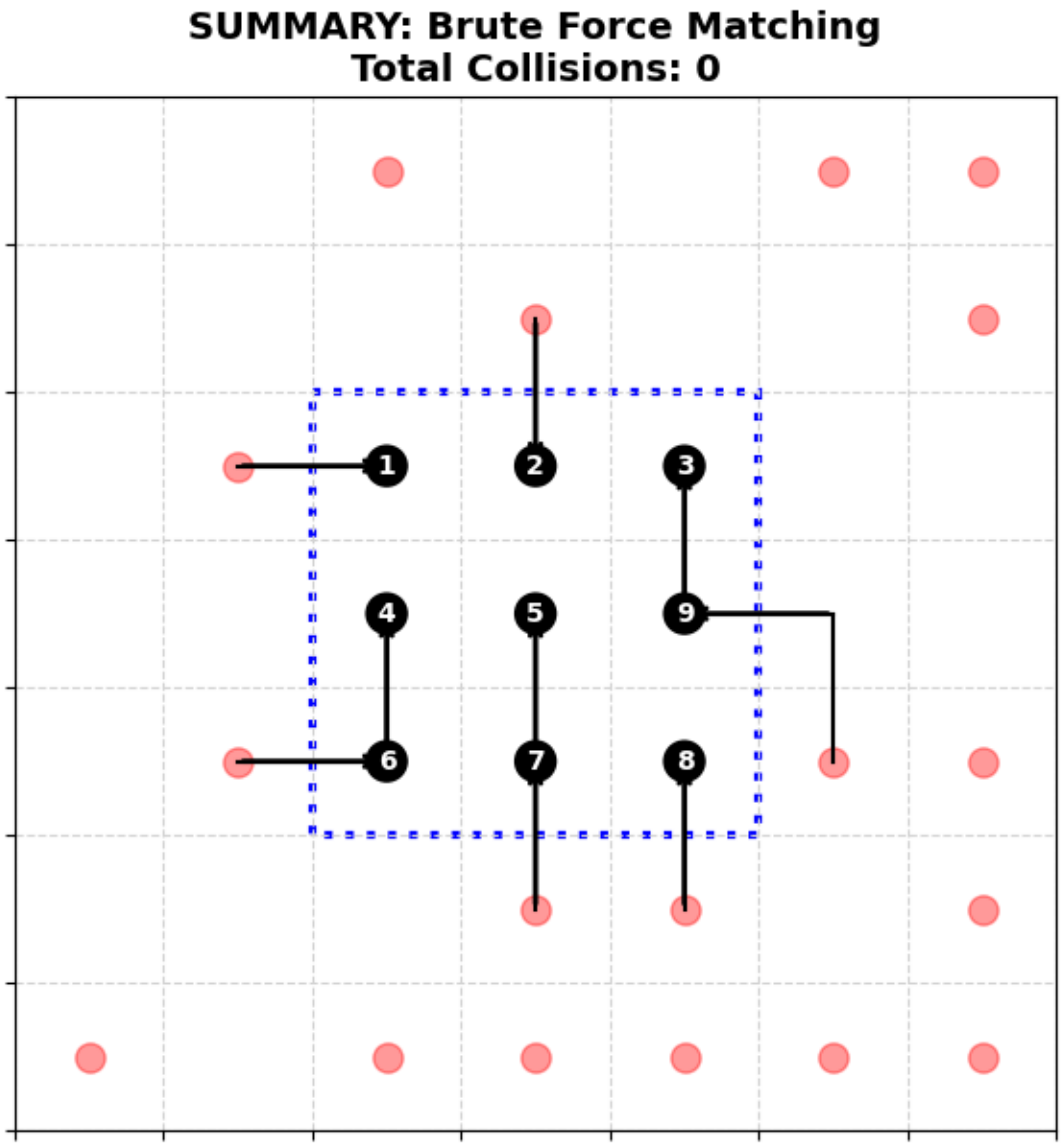


Paths

Evaluation Metrics

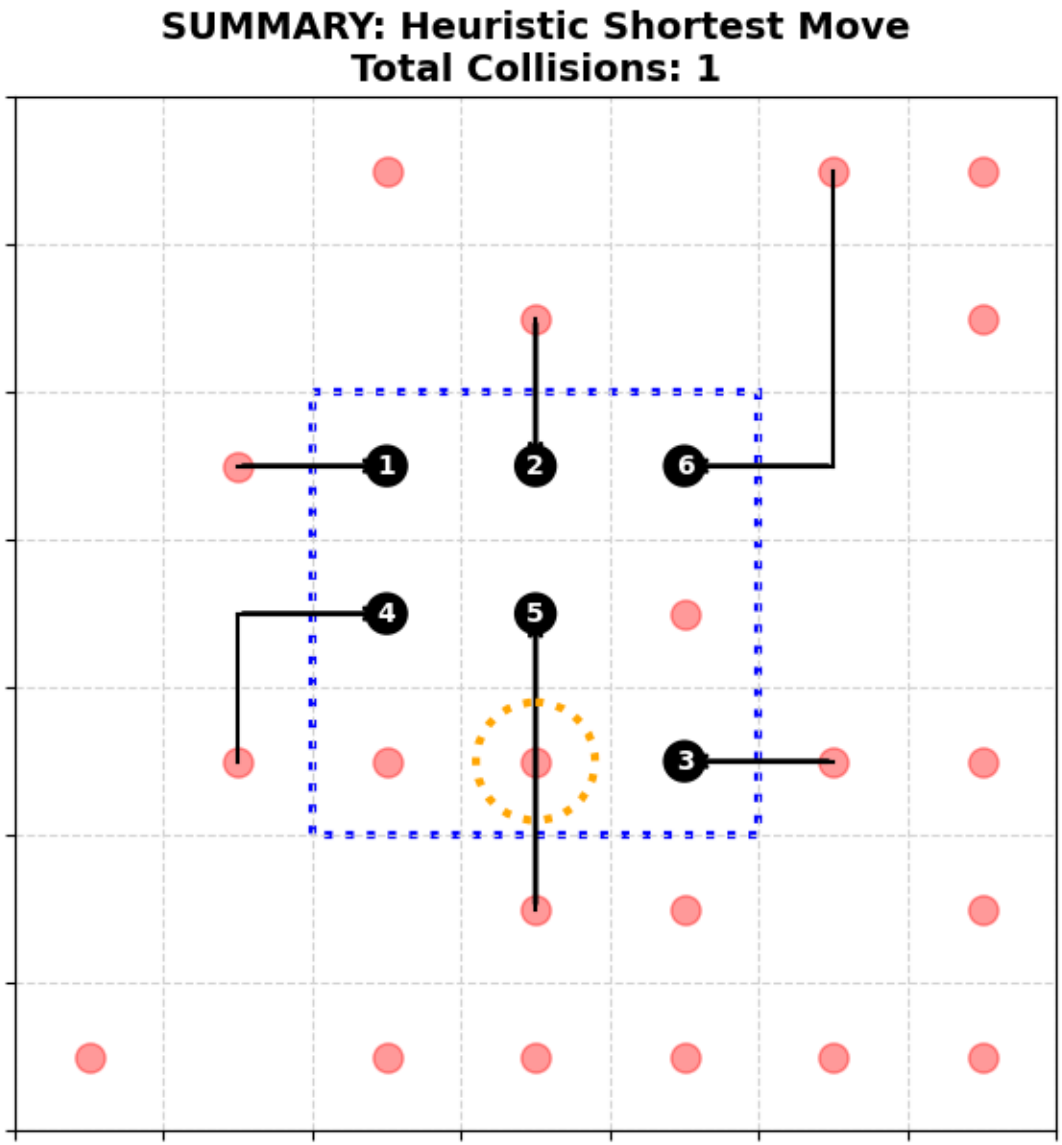
Semi Brute Force

- The program calculates distances for every combination of atoms and targets.
- To avoid crashing from factorial explosion, it dynamically limits the pool of candidate atoms.
- Only atoms closest to the target zone are considered, plus an additional two extra atoms.



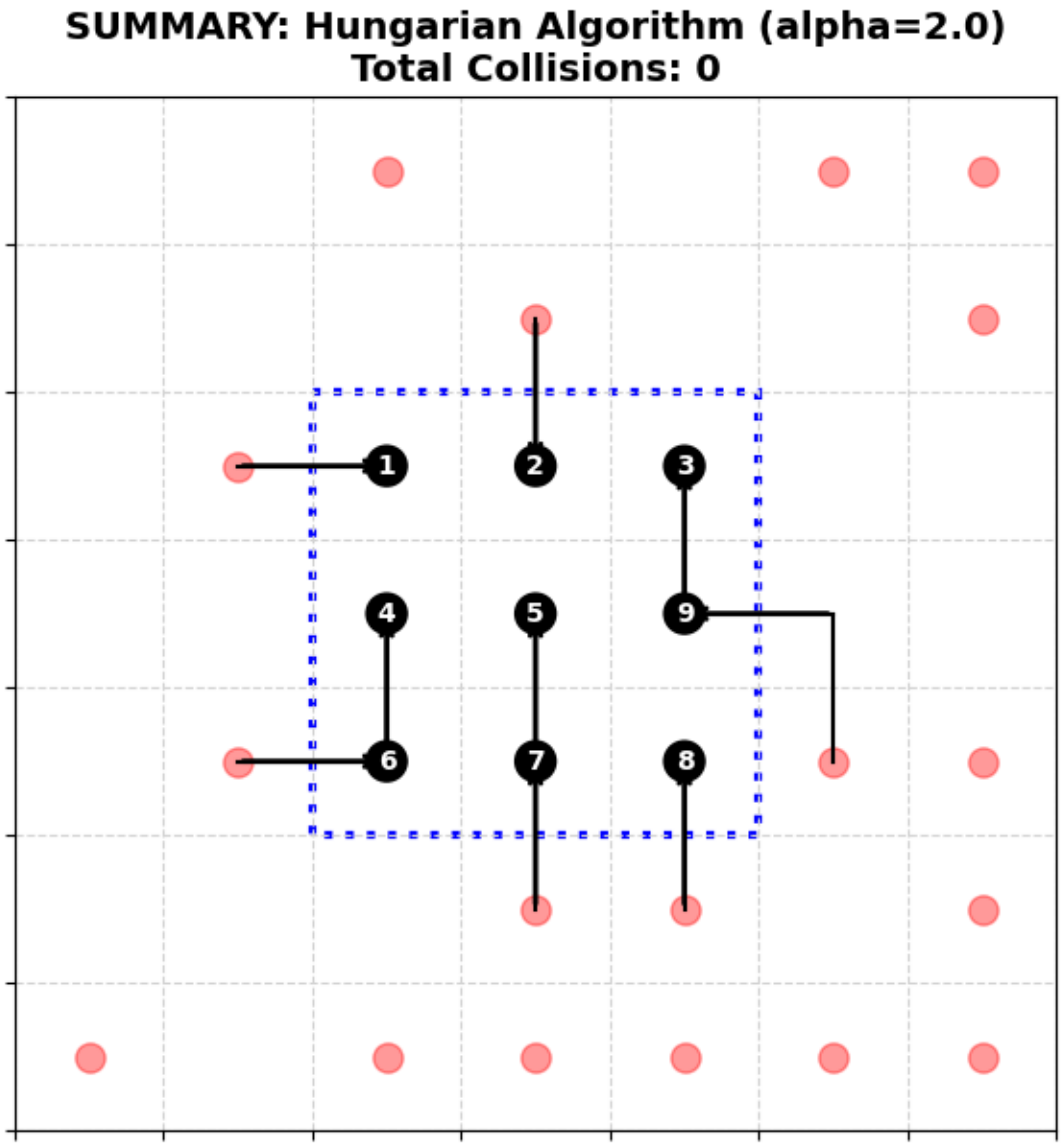
Heuristic Shortest Move

- Scans the entire board.
- Matches the closest atom to a target.
- Locks in the matched pairs.
- Removes the matched atoms.
- Repeats the process with the next closest pair until all targets are assigned.

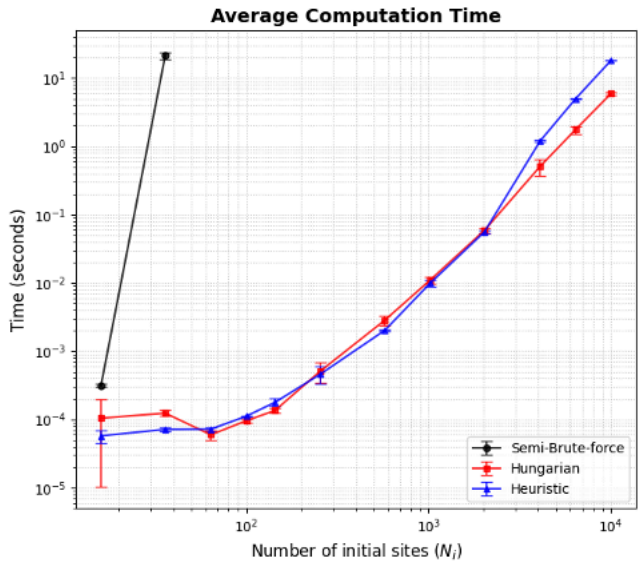


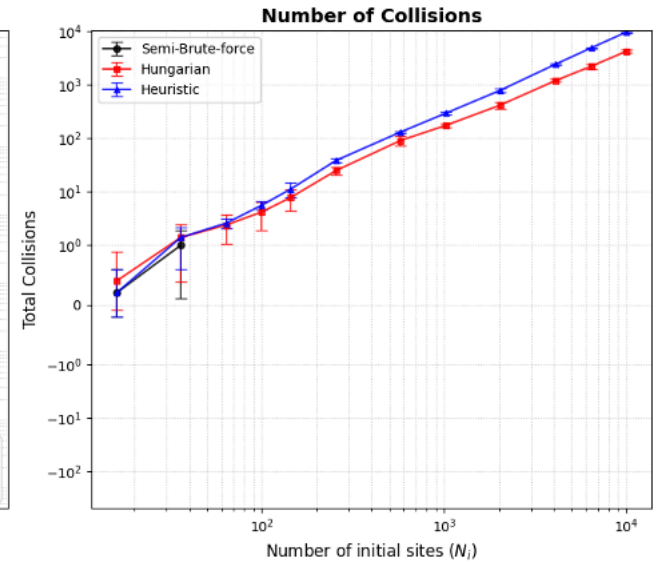
Hungarian Algorithm

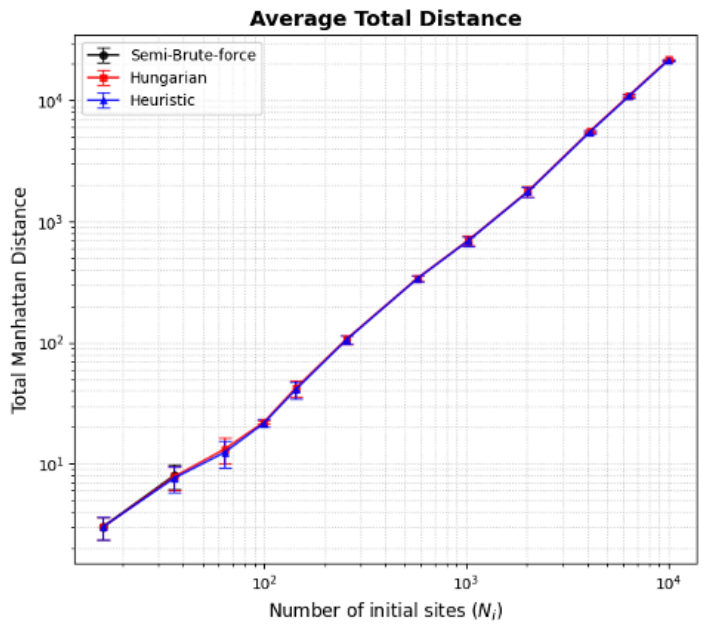
- The Hungarian algorithm avoids making greedy local choices.
- It focuses on the overall objective to minimize the total global distance traveled.
- The algorithm considers the combined movement of all atoms.



Results: Scaling Curves







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https://www.quantum-machines.co/solutions/neutral-atoms/

Thank you for listening!

Any questions?